C++ RECORD PROGRAM

University Management System Using Encapsulation, Constructors and Destructors

Aim

To design and implement a simple student management system using a C++ class with encapsulation, a constructor and a destructor, and to demonstrate the object lifecycle.

Algorithm

1. Start the program.
2. Define a Student class with private data members rollNo and name.
3. Use a constructor to initialize the student details.
4. Create a display() member function to show the details.
5. Create a Student object in main() and call display().
6. When the object goes out of scope, the destructor is called automatically.
7. End the program.

Program

```
#include <iostream>
#include <string>
using namespace std;

class Student
{
private:
    int rollNo;
    string name;

public:
    // Constructor
    Student(int r, string n)
    {
        rollNo = r;
        name = n;
        cout << "Student object created" << endl;
    }

    // Member function
    void display()
    {
        cout << "Roll Number: " << rollNo << endl;
        cout << "Name: " << name << endl;
    }

    // Destructor
    ~Student()
    {
        cout << "Student object destroyed" << endl;
    }
};

int main()
{
    Student s1(101, "Arun");
    s1.display();
    return 0;
}
```

Output

```
Student object created
Roll Number: 101
```

```
Name: Arun
Student object destroyed
```

Inference

The program demonstrates encapsulation by keeping the data members private and accessing them through a public member function. The constructor initializes the object when it is created, and the destructor is called automatically when the object is destroyed.

Result

Thus, a simple student management program was implemented successfully using encapsulation, a constructor and a destructor in C++, demonstrating the lifecycle of an object.